

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1918

Unionville 100 kV
November 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1918. The Interconnection Customer proposes a 64.3 MW battery energy storage facility interconnecting at the Unionville 100 kV substation in the Ohio area. The request was received on October 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 675 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 36 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$85,215,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ft Garland - Briarcliff Substation 100kV	Imperial - Eufaula 500kV	265.0	138	6.7
119Th St - Briarcliff 100kV	Imperial Substation - Eufaula 500kV L	253.8	132	6.7
Ft Garland - Beach Grove Substation 100kV	Eufaula - Imperial 500kV line	209.3	146	5.7
Poust - Fox 100kV	Imperial - Eufaula 500kV	201.3	132	5.2
Unionville - Venture 100kV	Eufaula - Beaver 500kV	190.3	149	19.1
Unionville - Venture 100kV Cir. 1	Eufaula - Beaver Substation 500kV	184.1	154	19.1
Beloit Gateway - Sierraville 100kV	Beaver - Eufaula 500kV	157.5	135	10.7
Fashion - Getty Substation 100kV L	Fashion Substation - Getty 100kV Circuit 1	155.9	157	5.1
Fashion Substation - Abbe 100kV L	Fashion - Abbe 100kV	153.6	179	5.6
Mesa Hts Substation - Natoma 230kV Line	Eufaula Substation - Beaver 500kV	151.6	373	7.5
John P Madgett - Beloit Gateway 100kV	Beaver - Eufaula 500kV L	150.2	131	10.7
Fashion - Getty 100kV	Getty - Fashion 100kV	147.6	165	5.1
Fashion - Abbe 100kV Cir. 1	Fashion - Abbe 100kV Circuit 2	144.9	190	5.6
Onancock Diesel Generation Facility - Mesa Hts 230kV	Beaver - Eufaula 500kV	137.2	442	7.5
Loring Power Plant - W Union 100kV	Imperial - Eufaula 500kV	129.4	197	5.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ntoma - Eufaula 230kV L	Beaver - Eufaula 500kV	127.9	424	7.5
Sierraville - Venture 100kV Line 1	Beaver - Eufaula 500kV	126.4	147	8.3
Venture - Sierraville 100kV Ckt 2	Eufaula - Beaver 500kV	126.4	147	8.3
Gallatin Tn - Don Julio-Lorac Vista 100kV	Beaver - Eufaula 500kV Line	124.7	141	14.6
Rock Rapids - Idylwild 100kV	Beaver - Trego Substation 500kV Line	122.2	295	6.7
Don Julio-Lorac Vista - Venture 100kV L	Eufaula - Bever 500kV	121.0	152	21.6
Beach Grove - Dollarhide 100kV	Beaver - Eufaula 500kV line	120.3	219	12.9
Idylwild - Ripp 100kV Line	Beaver - Trego 500kV line	119.1	326	6.7
Unionville - Beaver 230kV	Beaver - Eufaula 500kV Line	117.9	593	49.2
Axtel - Ripp 100kV	Trego - Beaver 500kV	116.8	356	6.7
Unionville 230-100 kV XFMR	Beaver - Eufaula 500kV	116.1	201	16.4
Unionville 230 kV/100 kV Auto #2	Beaver Substation - Eufaula 500kV	116.1	201	16.4
Unionville 230/100kV XFMR #3	Beaver - Eufaula 500kV	116.1	201	16.4
Rock Rapids - Firstenergy Edgewater 100kV	Trego - Beaver 500kV	114.4	164	13.0
Wiskeytown - E Franklin 100kV	Trego - Beaver 500kV Line	114.0	395	6.7
Baumhart - High Winds 100kV line	Beaver - Eufaula 500kV L	113.4	134	6.8
Dollarhide - John P Madgett 100kV	Beaver - Eufaula 500kV	111.9	376	16.5
Rock Rapids - High Winds 100kV Line	Beaver - Eufaula 500kV	110.7	138	6.8
Sierraville - Briarwood 100kV	Beaver - Eufaula 500kV	108.5	132	5.9
Gallatin Tn - Onancock Diesel Generation Facility 100kV Ckt 2	Unionville - Beaver 230kV	102.2	166	5.6
Onancock Diesel Generation Facility - Gallatin Tn 100kV Line	Beaver - Unionville 230kV	102.2	166	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ft Garland - Briarcliff Substation 100kV	Rebuild line with higher-capacity conductor	\$3,034,000	19% (\$566,000)
119Th St - Briarcliff 100kV	Rebuild line with higher-capacity conductor	\$1,000,000	19% (\$187,000)
Ft Garland - Beach Grove 100kV line	Rebuild line with higher-capacity conductor	\$3,450,000	10% (\$360,000)
Fox - Poust 100kV	Rebuild line with higher-capacity conductor	\$3,402,000	22% (\$756,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Unionville - Venture 100kV	Rebuild line with higher-capacity conductor	\$3,878,000	37% (\$1,428,000)
Unionville - Venture Substation 100kV Circuit 1	Rebuild line with higher-capacity conductor	\$3,259,000	37% (\$1,200,000)
Beloit Gateway Substation - Sierraville 100kV	Rebuild line with higher-capacity conductor	\$5,181,000	13% (\$689,000)
Fashion - Getty Substation 100kV L	Rebuild line with higher-capacity conductor	\$3,870,000	100% (\$3,870,000)
Fashion Substation - Abbe 100kV L	Rebuild line with higher-capacity conductor	\$3,836,000	100% (\$3,836,000)
Mesa Hts Substation - Natoma 230kV Line	Rebuild line with higher-capacity conductor	\$56,531,000	32% (\$18,082,000)
John P Madgett - Beloit Gateway Substation 100kV	Rebuild line with higher-capacity conductor	\$2,591,000	13% (\$344,000)
Fashion - Getty 100kV Line 1	Rebuild line with higher-capacity conductor	\$4,617,000	100% (\$4,617,000)
Fashion - Abbe 100kV #1	Rebuild line with higher-capacity conductor	\$2,037,000	100% (\$2,037,000)
Onancock Diesel Generation Facility - Mesa Hts 230kV	Rebuild line with higher-capacity conductor	\$20,633,000	32% (\$6,600,000)
Loring Power Plant - W Union 100kV	Rebuild line with higher-capacity conductor	\$1,685,000	26% (\$429,000)
Natma - Eufaula 230kV	Rebuild line with higher-capacity conductor	\$48,684,000	32% (\$15,572,000)
Sierraville - Venture 100kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,303,000	13% (\$306,000)
Venture - Sierraville 100kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,501,000	13% (\$200,000)
Gallatin Tn - Don Julio-Lorac Vista 100kV	Rebuild line with higher-capacity conductor	\$2,779,000	27% (\$741,000)
Rock Rapids - Idylwild 100kV	Rebuild line with higher-capacity conductor	\$2,138,000	8% (\$175,000)
Don Julio-Lorac Vista - Venture 100kV L	Rebuild line with higher-capacity conductor	\$5,543,000	34% (\$1,874,000)
Beach Grove Substation - Dollarhide 100kV	Rebuild line with higher-capacity conductor	\$2,377,000	9% (\$223,000)
Ripp - Idylwild 100kV	Rebuild line with higher-capacity conductor	\$2,426,000	8% (\$199,000)
Unionville Substation - Beaver 230kV L	Rebuild line with higher-capacity conductor	\$10,831,000	38% (\$4,157,000)
Axtel - Ripp 100kV line	Rebuild line with higher-capacity conductor	\$2,372,000	8% (\$194,000)
Unionville 230-100 kV XFMR	Replace transformer with higher-rated unit	\$9,520,000	47% (\$4,490,000)
Unionville 230 kV/100 kV Transformer T2	Replace transformer with higher-rated unit	\$11,190,000	47% (\$5,278,000)
Unionville 230 kV/100 kV TX T3	Replace transformer with higher-rated unit	\$7,976,000	47% (\$3,762,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Firstenergy Edgewater - Rock Rapids 100kV	Rebuild line with higher-capacity conductor	\$3,906,000	22% (\$863,000)
Wiskeytown - E Franklin 100kV	Rebuild line with higher-capacity conductor	\$3,273,000	8% (\$268,000)
Baumhart - High Winds 100kV line	Rebuild line with higher-capacity conductor	\$1,653,000	18% (\$302,000)
John P Madgett - Dollarhide 100kV	Rebuild line with higher-capacity conductor	\$1,908,000	9% (\$165,000)
Rock Rapids - High Winds 100kV Line	Rebuild line with higher-capacity conductor	\$2,193,000	18% (\$400,000)
Sierraville Substation - Briarwood 100kV	Reconductor line	\$2,098,000	13% (\$279,000)
Gallatin Tn - Onancock Diesel Generation Facility 100kV Ckt 2	Reconductor line	\$390,000	100% (\$390,000)
Gallatin Tn - Onancock Diesel Generation Facility 100kV Line	Reconductor line	\$376,000	100% (\$376,000)
	Total allocated to project		\$85,215,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.